✍️ Question (1): Definitions

Q1) Define Code Review.

Answer:
Code review is a systematic examination of source code performed by someone other than the original author to improve code quality, detect bugs, and help programmers learn from each other.

Q2) What are the purposes of Code Review?

Answer:
The purposes of code review are improving the code by finding bugs and improving clarity, and improving the programmer by sharing knowledge, techniques, and coding standards.

Q3) What are Coding Style Standards?

Answer:
Coding style standards are guidelines that define how code should be written to ensure readability, consistency, maintainability, and safety from bugs.

Q4) Define the DRY principle.

Answer:
DRY (Don't Repeat Yourself) is a principle that states that duplicated code should be eliminated by placing repeated logic into reusable components such as methods or classes.

Q5) What is meant by Fail Fast?

Answer:
Fail fast is a design principle that encourages detecting and reporting errors as early as possible to make debugging easier and prevent incorrect results.

Q6) What are Magic Numbers?

Answer:
Magic numbers are unexplained numeric constants in code that reduce readability and increase the risk of errors.

Q7) What is meant by One Purpose for Each Variable?

Answer:
Each variable should represent a single concept and should not be reused for different purposes to avoid confusion and bugs.

Q8) What are Global Variables and why should they be avoided?

Answer:
Global variables are variables that can be accessed and modified from anywhere in the program. They should be avoided because they make code harder to understand, debug, and maintain.

Q9) Why should methods return results instead of printing them?

Answer:
Methods should return results instead of printing them to make the code more reusable, flexible, and ready for change.

💻 Question (2): Code Review Question

Consider the following code snippet and answer the questions below:

نسخ الكود java

```java
public static int dayOfYear(int day, int month) {
    int result = day;
    if (month > 1) result += 31;
    if (month > 2) result += 28;
    if (month > 3) result += 31;
    if (month > 4) result += 30;
    return result;
}
```

Q1) Does the code violate the DRY principle? Explain.

Answer:
Yes, the code violates the DRY principle because similar logic is repeated multiple times using repeated conditional statements.

Q2) Does the code contain magic numbers?

Answer:
Yes, the code contains magic numbers such as 31, 28, and 30 which reduce readability and should be replaced with named constants or a data structure.

Q3) Is the code Fail Fast?

Answer:
No, the code does not fail fast because invalid inputs (such as incorrect month values) are not checked, which may lead to incorrect results.

Q4) Are the variable names appropriate?

Answer:
Partially. The variable name result is acceptable, but reusing the parameter day as part of another calculation can reduce clarity.

Q5) Does the method follow the rule "One purpose for each variable"?

Answer:

No, because the parameter day is reused to represent a different concept, which may confuse readers.

Q6) Is the method ready for change? Why?

Answer:

No, the method is not ready for change because modifying month lengths requires changing multiple hardcoded values in different places.
